Category Theory for Mathematical Physics — Exercise Sheet 7

Braiding: hexagons, Yang–Baxter and topological spin

Variant: Solutions

Conventions as in Lecture 7: the crossing $c_{X,Y}$ takes the strand from bottom left to top right over the other; $c_{a,b}\psi_c^{ab} = R_c^{ab}\,\psi_c^{ba}$ (Equation (7.5)); the hexagon equations are the two families of Theorem 7.5, with labels (a, b, c, d, e, g). The F-symbols are those of Lecture 6 (Fibonacci $1, \tau$; Ising $1, \sigma, \psi$, with the bare ψ Ising's fermion). P is the flip of two tensor factors. In Exercise 4, $\mathbb{Z}_2\times\mathbb{Z}_2$ is written additively, with the house names $1 = (0,0)$, $e = (1,0)$, $m = (0,1)$, $\varepsilon = (1,1)$ (here e is an anyon and ε the toric-code fermion); elsewhere $\mathbb{Z}_2 = \{1, s\}$ is multiplicative. $\varphi = (1+\sqrt{5})/2$. The braid generators σ_i never meet the Ising anyon σ; in Exercise 5 the unknown braiding is $c_{s,s} = z$, to keep c for a hexagon label; in Exercise 6 the unknown $R_\sigma^{\sigma\psi}$ is named y (A is reserved for $q = A^2$).

Exercise 1 (Braids and permutations). (i) Show that $\sigma_i \mapsto (i\ i{+}1)$ defines a surjective homomorphism $B_n \to S_n$, and $\sigma_i \mapsto 1$ a homomorphism $B_n \to \mathbb{Z}$ (with $\mathbb{Z}$ additive). Conclude that for $n \geq 2$, σ_1 has infinite order in B_n, while its image in S_n has order 2.

(ii) For $n = 3$ draw $\sigma_1\sigma_2\sigma_1$, $\sigma_2\sigma_1\sigma_2$, and σ_1^2. Which drawing shows why $\sigma_1^2 \neq 1$?

Solution. (i) By § 7.1, B_n is generated by $\sigma_1, \dots, \sigma_{n-1}$ with the relations $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ and $\sigma_i\sigma_j = \sigma_j\sigma_i$ for $|i-j| \geq 2$, so a map of the generators into a group extends to a homomorphism exactly when the images satisfy them. The transpositions do: $(i\ i{+}1)(i{+}1\ i{+}2)(i\ i{+}1) = (i\ i{+}2) = (i{+}1\ i{+}2)(i\ i{+}1)(i{+}1\ i{+}2)$, and transpositions of disjoint pairs commute. They generate S_n, so the map is onto. For $\sigma_i \mapsto 1 \in \mathbb{Z}$ each relation has the same number of letters on both sides ($3 = 3$, $2 = 2$), so the images agree. The image of σ_1^k in $\mathbb{Z}$ is k, so $\sigma_1^k \neq 1$ for $k \neq 0$: σ_1 has infinite order. Its image $(1\,2)$ has order 2. In particular $B_n \to S_n$ is not injective: σ_1^2 is in its kernel.

(ii) $\sigma_1\sigma_2\sigma_1$ and $\sigma_2\sigma_1\sigma_2$ are the two sides of Equation (7.4): the same three crossings, the strand from the left passing over the other two, in a different order; Reidemeister III moves one into the other. σ_1^2 is the left side of Equation (7.3), two crossings of the same kind: the first strand goes once around the second. Its permutation is the identity, but the picture cannot be undone without passing a strand through another, which a braid in the plane may not do; that is what the map to $\mathbb{Z}$ detects.

Exercise 2 (Two forms of Yang–Baxter). (i) Draw both sides of the braid relation of Theorem 7.4 for strands X, Y, Z, and say which steps of its proof use the first hexagon and which use naturality.

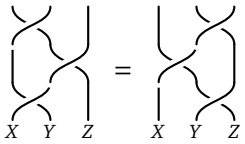
(ii) Let R act on $V \otimes V$, let $c = P \circ R$, and write R_{12}, R_{13}, R_{23} and c_{12}, c_{23} for the operators on $V^{\otimes 3}$. Show that

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23} \iff R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

(iii) The braiding of sVec on $V = \mathbb{C}^{1|1}$ is $c(v \otimes w) = (-1)^{|v||w|} w \otimes v$. Find its R, check both forms, and check that $c^2 = 1$.

Solution. (i) The proof of Theorem 7.4, drawn. Draw X, Y, Z from left to right. The left side has X crossing over Y, then over Z, then Y crossing over Z. By the first hexagon (Equation (7.1)) the first two crossings are the single crossing of X over the pair $Y \otimes Z$. The

crossing of Y over Z, a morphism on the pair, slides through that crossing to its other side (naturality, Equation (7.2)): now Y crosses Z first, then X crosses over the pair $Z \otimes Y$. The first hexagon again splits this into X over Z, then X over Y: the right side. Read as a picture, the strand X has slid over the crossing of Y and Z, which is Equation (7.4):



The first two crossings on the left (bottom) are, by the hexagon, X crossing the pair $Y \otimes Z$; the crossing of Y and Z on top slides down through it.

(ii) Write X_{ab} for X acting in tensor factors a and b. Conjugating by a flip relabels the factors: $P_{12}X_{23}P_{12} = X_{13}$, $P_{23}X_{12}P_{23} = X_{13}$, $P_{12}X_{13}P_{12} = X_{23}$, $P_{23}X_{13}P_{23} = X_{12}$. With $c_{12} = P_{12}R_{12}$ and $c_{23} = P_{23}R_{23}$,

$$\begin{aligned} c_{12}c_{23}c_{12} &= P_{12}R_{12}P_{23}R_{23}P_{12}R_{12} \\ &= P_{12}P_{23}(P_{23}R_{12}P_{23})R_{23}P_{12}R_{12} \\ &= P_{12}P_{23}R_{13}R_{23}P_{12}R_{12} \\ &= P_{12}P_{23}P_{12}(P_{12}R_{13}R_{23}P_{12})R_{12} \\ &= P_{12}P_{23}P_{12}\,R_{23}R_{13}R_{12}, \end{aligned}$$

using $P^2 = 1$. In the same way $c_{23}c_{12}c_{23} = P_{23}P_{12}P_{23}\,R_{12}R_{13}R_{23}$. Both permutations $P_{12}P_{23}P_{12}$ and $P_{23}P_{12}P_{23}$ exchange the first and third factors, so they are equal and invertible; cancelling them gives the equivalence.

(iii) In the basis $00, 01, 10, 11$ (degree 0 first), $c = P\,\mathrm{diag}(1,1,1,-1)$, since only $1 \otimes 1$, odd with odd, gets the sign: $R = P^{-1}c = \mathrm{diag}(1,1,1,-1)$. Then R_{12}, R_{13}, R_{23} are diagonal in the product basis, so they commute and $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$; by (ii) c satisfies the braid form. And $c^2(v \otimes w) = (-1)^{|v||w|}(-1)^{|w||v|}v \otimes w = v \otimes w$: the braiding is symmetric, as Example 7.3 says. (The script also checks both forms for the B_q of Lecture 8.)

Exercise 3 (B_3 on three τ's). Take Fibonacci with $R_1^{\tau\tau} = e^{-4\pi i/5}$ and $R_\tau^{\tau\tau} = e^{3\pi i/5}$ (Theorem 7.6), and $F = F_\tau^{\tau\tau\tau}$ of Theorem 6.6.

(i) On $V_\tau^{\tau\tau\tau}$, with the basis of left trees L_1, L_τ, show that σ_1 acts by $M_1 = \mathrm{diag}(R_1^{\tau\tau}, R_\tau^{\tau\tau})$ and σ_2 by $M_2 = FM_1F$.

(ii) Show that the braid relation $M_1M_2M_1 = M_2M_1M_2$ is equivalent to $(FM_1)^3$ commuting with F, and check that $(FM_1)^3$ is a multiple of the identity.

(iii) How do σ_1 and σ_2 act on $V_1^{\tau\tau\tau}$?

Solution. (i) By Equation (7.5), σ_1, the crossing of the first two strands, acts on the left tree $L_e = (\psi_e^{\tau\tau} \otimes \mathrm{id}) \circ \psi_\tau^{e\tau}$ by $R_e^{\tau\tau}$: $M_1 = \mathrm{diag}(R_1^{\tau\tau}, R_\tau^{\tau\tau})$. For σ_2, write L_e in right trees, $L_e = \sum_f F_{ef}R_f$ (Equation (6.1), the associator being invisible); σ_2 multiplies R_f by $R_f^{\tau\tau}$; and $R_f = \sum_{e'} \overline{F_{e'f}}L_{e'}$ (Equation (6.3)). So $\sigma_2 L_e = \sum_{e'}\big(\sum_f \overline{F_{e'f}}R_f^{\tau\tau}F_{ef}\big)L_{e'}$. As F is real and symmetric, the coefficient is $(FM_1F)_{e'e}$: $M_2 = FM_1F$.

(ii) With $F^2 = 1$, $M_1M_2M_1 = M_1FM_1FM_1$ and $M_2M_1M_2 = FM_1FM_1FM_1F$. Multiplying the equation $M_1FM_1FM_1 = FM_1FM_1FM_1F$ on the left by F gives $(FM_1)^3 = F(FM_1)^3F$, and conversely: the braid relation holds exactly when $(FM_1)^3$ commutes with F. Computing (with $\varphi^2 = \varphi + 1$, or numerically), $(FM_1)^3 = e^{-4\pi i/5}\,1 = R_1^{\tau\tau}\,1$, a scalar, so it commutes with F and the braid relation holds.

(iii) In $V_1^{\tau\tau\tau}$ the first two τ's must fuse to τ (then $\tau \otimes \tau \to 1$), and the same for the last two: the space is a line, and both generators act by $R_\tau^{\tau\tau}$. The relation $(R_\tau^{\tau\tau})^3 = (R_\tau^{\tau\tau})^3$ holds trivially.

Exercise 4 (The toric code as a braided category). Take $\mathrm{Vec}_{\mathbb{Z}_2\times\mathbb{Z}_2}$ with trivial associator, and a braiding acting on simple objects by numbers, $c_{x,y} = \beta(x,y)\,\mathrm{id}$, with $\beta(x,0) = \beta(0,y) = 1$.

(i) Show that both families of hexagon equations hold exactly when β is a *bicharacter*: $\beta(x+y,z) = \beta(x,z)\beta(y,z)$ and $\beta(x,y+z) = \beta(x,y)\beta(x,z)$.

(ii) Let $\beta(x,y) = (-1)^{x_1y_2}$. Compute the spins θ_x of $1, e, m, \varepsilon$ and the table of double braidings $\beta(x,y)\beta(y,x)$. Is $\beta(x,y) = \beta(y,x)$? Is the braiding symmetric in the sense of Definition 7.2?

(iii) Show that $\beta'(x,y) = (-1)^{x_2y_1}$ is the gauge transform of β by $u(x,y) = \beta(x,y)$, which keeps the trivial associator.

Solution. (i) The category is pointed: every $x \otimes y$ is the simple $x+y$, and every F-symbol is 1 (trivial associator). In the first family with outer labels a, b, c every channel is forced (the total charge is $a+b+c$), and the sum has the single term $f = a+b$; the equation reads $R^{ca}R^{cb} = R^{c,a+b}$, that is $\beta(c,a)\beta(c,b) = \beta(c,a+b)$: multiplicativity in the second argument. The second family reads $(R^{ac})^{-1}(R^{bc})^{-1} = (R^{a+b,c})^{-1}$, multiplicativity in the first. So both hold for all labels exactly when β is a bicharacter.

(ii) $\beta(x,y) = (-1)^{x_1y_2}$ is a bicharacter (the exponent is bilinear mod 2), so it is a braiding. By Theorem 7.7 with $d = 1$, $\theta_x = R^{xx} = \beta(x,x) = (-1)^{x_1x_2}$: $1, 1, 1, -1$ for $1, e, m, \varepsilon$; ε is a fermion. The double braiding is $(-1)^{x_1y_2+y_1x_2}$: it is -1 for (e,m), (e,ε) and (m,ε) (in either order) and 1 for every other pair. Taking e around m gives -1, the phase *Quantum Computing* measured (Equation (7.8)). The function β is not symmetric in its arguments: $\beta(e,m) = (-1)^{1\cdot1} = -1$ but $\beta(m,e) = (-1)^{0\cdot0} = 1$. Nor is the braiding symmetric in the sense of Definition 7.2, since its double braiding is not always 1.

(iii) A gauge with vertex rescalings $u(x,y) = u_{x+y}^{xy}$ multiplies R^{xy} by $u(y,x)/u(x,y)$ (§ 7.4) and changes the associator ω by the coboundary of u (Lecture 6); for a bicharacter u, $\delta u = 1$, so ω stays trivial. With $u = \beta$: $\beta(x,y)\,\beta(y,x)/\beta(x,y) = \beta(y,x) = (-1)^{y_1x_2} = \beta'(x,y)$. So β and β' are the same braided category, with the same spins, as the classification by quadratic forms $q(x) = \theta_x$ (§ 7.6, cited) requires.

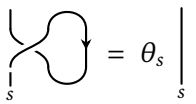
Exercise 5 (Spins: the semion and Fibonacci). (i) For Vec_G^ω on $\mathbb{Z}_2$ with $\omega(s,s,s) = -1$ and $c_{s,s} = z$, write the instance $a = b = c = d = s$, $e = g = 1$ of the second hexagon family, and show with Theorem 7.8 that $z = \pm i$. Draw the curl that defines θ_s (Equation (7.7)) and compute θ_s and the double braiding for the semion $z = i$.

(ii) For Fibonacci with $\theta_\tau = e^{4\pi i/5}$, check the ribbon relation $R_c^{\tau\tau}R_c^{\tau\tau} = \theta_c/\theta_\tau^2$ for $c = 1, \tau$. Is τ transparent?

Solution. (i) With $a = b = c = d = s$ and $e = g = 1$ the second family reads

$$\begin{aligned}&(R_1^{ss})^{-1}[F_s^{sss}]_{11}(R_1^{ss})^{-1}\\ &\quad= [F_s^{sss}]_{11}(R_s^{1s})^{-1}[F_s^{sss}]_{11},\end{aligned}$$

the sum having the single term $f = 1$: $z^{-1}\omega z^{-1} = \omega^2$, so $z^2 = \omega^{-1} = \omega$. With $\omega = -1$ this is $z^2 = -1$, $z = \pm i$, as the first family gave in Theorem 7.8. The curl of Equation (7.7) is the strand s crossing over itself and closed on the right:



By Theorem 7.7 with $d_s = 1$ and the single channel $s \otimes s \cong 1$, $\theta_s = R_1^{ss} = z = i$ for the semion, and the double braiding is $z^2 = -1$: two semions exchanged twice pick up -1, which no boson or fermion does. The antisemion $z = -i$ has $\theta_s = -i$; as θ_s is invariant, the two are different braided categories.

(ii) $\theta_1 = 1$ and $\theta_\tau^2 = e^{8\pi i/5}$. For $c = 1$: $(R_1^{\tau\tau})^2 = e^{-8\pi i/5} = \theta_1/\theta_\tau^2$. For $c = \tau$: $(R_\tau^{\tau\tau})^2 = e^{6\pi i/5}$ and $\theta_\tau/\theta_\tau^2 = \theta_\tau^{-1} = e^{-4\pi i/5} = e^{6\pi i/5}$. Both hold. The double braiding of τ with itself acts on the channel c by $R_c^{\tau\tau}R_c^{\tau\tau}$ (Equation (7.8)), which is $e^{-8\pi i/5} \neq 1$ and $e^{6\pi i/5} \neq 1$: τ is not transparent, and the Müger centre of Fibonacci is $\{1\}$.

Exercise 6 ⋆. *The Ising hexagons, solved.* Take Ising with the F-symbols of Theorem 6.7 for $\kappa = +1$ ($F_\sigma^{\sigma\sigma\sigma} = H$, $[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma} = [F_\sigma^{\psi\sigma\psi}]_{\sigma\sigma} = -1$, the rest 1 where the trees exist). The unknowns are $R_1^{\sigma\sigma}$, $R_\psi^{\sigma\sigma}$, $R_\sigma^{\sigma\psi}$, $R_\sigma^{\psi\sigma}$ and $R_1^{\psi\psi}$.

(i) Write out the first- and the second-family instances with labels $(a, b, c, d, e, g) = (\sigma, \psi, \sigma, \psi, 1, \sigma)$, and deduce $R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma}$.

(ii) Write out the first-family instances with labels $(\psi, \psi, \sigma, \sigma, \sigma, \sigma)$, $(\sigma, \sigma, \psi, 1, \sigma, \sigma)$ and $(\sigma, \sigma, \sigma, \sigma, 1, 1)$, and use them to show that there are at most four solutions. Which values of ν in $\theta_\sigma = e^{i\pi\nu/8}$ do they give?

(iii) Write out the solution with $\nu = 1$, and compute θ_ψ.

Solution. (i) With $(a, b, c, d, e, g) = (\sigma, \psi, \sigma, \psi, 1, \sigma)$ the first family is $R_e^{ca}[F_d^{acb}]_{eg}R_g^{cb} = \sum_f [F_d^{cab}]_{ef}R_d^{cf}[F_d^{abc}]_{fg}$, that is $R_1^{\sigma\sigma}[F_\psi^{\sigma\sigma\psi}]_{1\sigma}R_\sigma^{\sigma\psi} = \sum_f [F_\psi^{\sigma\sigma\psi}]_{1f}R_\psi^{\sigma f}[F_\psi^{\sigma\psi\sigma}]_{f\sigma}$. The trees of $F_\psi^{\sigma\sigma\psi}$ with left channel 1 need $f = \sigma$ (the channel of $\sigma \otimes \psi$), so the sum has one term: $R_1^{\sigma\sigma} \cdot 1 \cdot R_\sigma^{\sigma\psi} = 1 \cdot R_\psi^{\sigma\sigma} \cdot (-1)$. The second family at the same labels exchanges the labels of the inverted crossings:

$$(R_1^{\sigma\sigma})^{-1} \cdot 1 \cdot (R_\sigma^{\psi\sigma})^{-1} = 1 \cdot (R_\psi^{\sigma\sigma})^{-1} \cdot (-1),$$

that is $R_\psi^{\sigma\sigma} = -R_\sigma^{\psi\sigma}R_1^{\sigma\sigma}$. As $R_1^{\sigma\sigma} \neq 0$, $R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma}$; call it y.

(ii) $(\psi, \psi, \sigma, \sigma, \sigma, \sigma)$: the left side is $R_\sigma^{\sigma\psi}[F_\sigma^{\psi\sigma\psi}]_{\sigma\sigma}R_\sigma^{\sigma\psi} = -y^2$, and the right side the single term $f = 1$, $[F_\sigma^{\sigma\psi\psi}]_{\sigma 1}R_\sigma^{\sigma 1}[F_\sigma^{\psi\psi\sigma}]_{1\sigma} = 1$: $y^2 = -1$, so $y = \pm i$. $(\sigma, \sigma, \psi, 1, \sigma, \sigma)$: the left side is $R_\sigma^{\psi\sigma} \cdot 1 \cdot R_\sigma^{\psi\sigma} = y^2$, the right side $R_1^{\psi\psi}$ (one term, $f = \psi$, F-symbols 1): $R_1^{\psi\psi} = y^2 = -1$, so ψ is a fermion. $(\sigma, \sigma, \sigma, \sigma, 1, 1)$: the left side is $R_1^{\sigma\sigma}\,[F_\sigma^{\sigma\sigma\sigma}]_{11}\,R_1^{\sigma\sigma} = (R_1^{\sigma\sigma})^2/\sqrt{2}$; on the right, $f = 1$ gives $\frac{1}{\sqrt{2}} \cdot 1 \cdot \frac{1}{\sqrt{2}}$ and $f = \psi$ gives $\frac{1}{\sqrt{2}}\,R_\sigma^{\sigma\psi}\,\frac{1}{\sqrt{2}}$: $\sqrt{2}\,(R_1^{\sigma\sigma})^2 = 1 + y = e^{\pm i\pi/4}\sqrt{2}$ for $y = \pm i$, so $R_1^{\sigma\sigma} = \pm e^{\pm i\pi/8}$, with the exponent's sign that of y/i, and $R_\psi^{\sigma\sigma} = -yR_1^{\sigma\sigma}$ by (i). That is at most four solutions, two for each y. Each is one: the script checks every instance of both families, and they are the four braidings of Theorem 7.9 with $\kappa = +1$. By Theorem 7.7, $\theta_\sigma = (R_1^{\sigma\sigma} + R_\psi^{\sigma\sigma})/\sqrt{2} = R_1^{\sigma\sigma}(1 - y)/\sqrt{2}$. For $y = -i$, $(1 + i)/\sqrt{2} = e^{i\pi/4}$, and $R_1^{\sigma\sigma} = \pm e^{-i\pi/8}$ gives $\theta_\sigma = \pm e^{i\pi/8}$: $\nu = 1$ or $\nu = 9$. For $y = i$, $(1 - i)/\sqrt{2} = e^{-i\pi/4}$, and $R_1^{\sigma\sigma} = \pm e^{i\pi/8}$ gives $\theta_\sigma = \pm e^{-i\pi/8}$: $\nu = 15$ or $\nu = 7$.

(iii) $\nu = 1$: $y = -i$ and $R_1^{\sigma\sigma} = e^{-i\pi/8}$, so $R_\psi^{\sigma\sigma} = ie^{-i\pi/8} = e^{3i\pi/8}$, $R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma} = -i$ and $R_1^{\psi\psi} = -1$: the data printed in § 7.6. Then $\theta_\psi = R_1^{\psi\psi} = -1$ (one channel, $d_\psi = 1$), and $\theta_\sigma = e^{i\pi/8}$.